PROVENANCE AND EVIDENCE SUPPLEMENT FOR HIGH-WINDING STATIONARY SOLUTIONS AND SINGULAR EXITS IN THE VAN DER POL REACTION–DIFFUSION SYSTEM

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1. Purpose and release identity

This supplement records evidence status, source identities, theorem dependencies, and replay information for the companion paper. It is not a second mathematical proof and does not enlarge the theorem. Two immutable public records have different roles in the argument. The publicly archived fixed-system preprint is Zenodo version 1.0.0,

`https://doi.org/10.5281/zenodo.22139860`,

whereas the version-1 application proof dossiers, certificates, and exact frozen long-draft snapshot are fixed together by

`https://github.com/h-lu/rfsn-ii-positive-parameter-pde/releases/tag/vdp-companion-v1`.

The two records are not interchangeable. The Zenodo paper is the stable source for the fixed-Hamiltonian high-winding theorem and its RFSN-II realization. The application release is the immutable evidence baseline for the older long-draft clauses not retained in that focused paper and for the coordinate-level version-1 positive-parameter arguments. The present reader-facing manuscript may be revised without changing that archived baseline. The read-only upstream repository was not modified.

The companion theorem is conditional on Hypothesis H. Its analytic application conclusions are existential on a nonempty compact positive annulus. The proposed numerical Issue #7 box is not used as a theorem box.

2. Public fixed-system source and frozen long source

2.1. **Public fixed-system preprint.** The stable fixed-system source is H. Lu, *High-winding first-hit geometry and singular action finite parts near a reversible Hamiltonian saddle-focus*, Zenodo version 1.0.0, DOI `https://doi.org/10.5281/zenodo.22139860`. Its source revision is

`3516a2ee499fc74a46d0ec204736bde8d3782c59`,

and the deposited manuscript PDF has SHA-256

`569906de86c7d3a716831e036056cfeebb772d5a87fb89c04ee61ee3daf0787d`.

Date: August 28, 2026.

That record contains the focused manuscript, its self-contained source, and the M1–M4 evidence used for the RFSN-II realization. In particular, it is the stable source for the fixed-system first-hit theorem, the RFSN-II core homoclinic and three selected channels, the fixed-system return and action modules, and the countable coding consequence; the principal locations are Theorems 2.9 and 6.3 and Corollaries 7.1–7.2. Its public replay summary records `PASS` for the M1–M4 `main-theorem` profile. This is a replay of the stated computer-assisted inputs, not an independent review of the analytic arguments.

The focused paper deliberately treats one fixed Hamiltonian. It does not state the compact-parameter, mixed state/parameter versions used by the application, and it does not contain the optional Jost module described below. Moreover, its concrete pole theorem concerns an isolated boundary sink with indicial roots $\{1, 4, 6\}$; it is not the positive-parameter van der Pol pole theorem.

2.2. **Frozen long research version.** The auxiliary frozen source is the earlier long research version H. Lu, *First returns, singular exits, and action finite parts near a reversible Hamiltonian saddle-focus*, originally stored in the historical repository `h-lu/reversible-rfsn-ii-waves`. It is not another name for the Zenodo paper. The inspectable copy is the exact snapshot inside the application release. Its upstream commit is

`d54add098545063d5efe8f1d6f062d4cfc116a0d`

and tree

`b9cd34b1bae1c29bdd722d9ed3c33402c4b3ee89`.

The public application release contains an exact `git archive` extraction of the complete frozen `papers/paper-a/manuscript/` and `validation/` directories at

`frozen-imports/rfsn-ii-d54add098545063d5efe8f1d6f062d4cfc116a0d/source/`.

No file below that directory was edited after extraction.

The principal SHA-256 identities of this long snapshot are:

frozen file	SHA-256
manuscript source	`0baf6335aad72d5893479d8876d2613671ecb8ac2ccd73664405dea4381e6a20`
manuscript PDF	`67888cf8b61b34c923cf55bd69ee41cab69493be6fc275533c8f5a074f1e96c5`
replay manifest	`15905f5a20b24a0ae0d298d9d14aa940177a006cc2cce3f2a39fd2e1cf4dac9b`
environment lock	`6240ebdbf0f296738534c07a33aea40202883f6abf37e1ff28e43dad47aa0cba`
core homoclinic certificate	`ed0f9f58f8ba5f1d5c36dc7c3a72bb725599c4172a3cd610d890b88699fecfbd`
algebraic-anchor certificate	`60882ee1d3b2b18264b85764288505ae8b47d00bc826a2bddec152898f690fbe`
future-target certificate	`88fa64035bb4352f5e25aa8d1627b191936264958c125dface59c5a767f6b3ce`
pole-gate certificate	`7f325a87810f8b0dda2542aed90263b39f9a9f4bd2e8ebb3abcc238d032eb6e2`

The machine-readable copy of this table is `RELEASE_MANIFEST.json` beside the frozen snapshot README.

The four displayed claim-bearing certificate files are byte-identical to the corresponding M1–M4 evidence in the Zenodo record. The long snapshot is kept because some analytic clauses used by the companion are absent from, or strictly stronger than, the focused paper; it is not used to replace the stable citation when the focused theorem already supplies the required fixed-system statement.

3. Imported clauses and exclusions

The following clauses of Hypothesis H are supplied directly by the public fixed-system preprint.

(1) The core system, one selected symmetric homoclinic, transversality modulo flow in the regular zero-energy hypersurface, its true unstable source graph, the certified channel anchors, and the adapted finite central physical event block.
(2) The fixed-system saddle passage, winding selector, whole-cell convergence, clean first-event decomposition, exact finite-branch action and covariance, and countable coding of the trapped return set.
(3) The C^3, third-order-bunched singular algebraic comparison sheet and the validated transverse source-to-sheet intersection. The nonzero source-incidence row in Hypothesis H is the corresponding section-level restatement of this bordered transversality.

Only the following additional clauses are taken from the frozen long version.

(1) The C^4, fourth-rate-gap form of the canonical singular algebraic comparison sheet used in the matching proof (Proposition 8.1 of the long source).
(2) The exact Jost basis and tangent identification, with the endpoint-normalized adjoint row whose directed exchange pairing is $144\sqrt{3}$. This coefficient is not a theorem statement in Zenodo version 1.0.0; it is the explicit algebraic consequence of the frozen Cauchy data and symplectic pairings in `app:jost-source-arm`, as recorded in the application proof archive.
(3) The compact-parameter architecture for the endpoint and matching equations, mixed passage and selector, mixed two-jet estimates, and finite-atlas covariance. The concrete compact-family hypotheses and the finite marked-atlas descent are checked in the companion paper, with coordinate-level details retained in the application archive; the long version does not by itself verify their van der Pol hypotheses.

Thus Hypothesis H combines a publicly archived fixed-system theorem with a small, explicitly identified set of long-version clauses. Its conditional status is a theorem-scope statement, not a claim that the focused flagship preprint is unarchived or inaccessible.

The following conclusions are not imported from either source:

- either concrete singular end as a positive-parameter van der Pol end;
- a positive-parameter pole or algebraic compactification;
- a positive-parameter $K_2 \to K_1 \to$ outer matching theorem;
- a positive-parameter terminal counterterm or finite part;

- a global exact saddle chart or one parameter-global winding alphabet;
- temporal PDE stability, temporal chaos, Turing selection, canard identification, or experiment.

Thus the two positive ends, matching bridge, finite parts, event refinements, and translation to the original PDE are application results, not imported conclusions of the fixed-system theory.

4. Claim register crosswalk

The status column below reproduces the authoritative register verbatim. The dependency column records where a proved application statement uses Hypothesis H or an earlier H-dependent statement. Thus *Proved* is a proof status, not a claim that the frozen certificates were independently replayed by this repository.

ID	statement	status	dependency
V1	physical reversible exact Hamiltonian structure and primitive	Derived	direct calculation
V2	central saddle-focus, selected homoclinic, exact passage, and finite events	Proved	public $\mathrm{H}_{\mathrm{core}}$; long/local family passage
V3	genuine positive pole, open basin, labels, and pole action finite part	Proved	V2 pole window
V4	positive outer future-staying hypersurface and algebraic asymptotics	Proved	direct full-PDE construction
V5	$K_2 \to K_1 \to$ outer matching and moving-cut covariance	Proved	V2, V4; public incidence and long Jost row
V5A	reference-subtracted algebraic length/action finite parts	Proved	matched V5 arrival
V6	exhaustive physical high-winding return/first-exit relation	Proved	V2–V5A; public fixed-system and long/local family modules
V7	periodic, multipulse, and aperiodic stationary PDE patterns	Proved	V6; public/local coding modules
V8	periodic centered-moment criterion for a positive co-periodic eigenvalue	Proved	direct temporal operator-pencil argument
V8A	frozen A2 true profile satisfies the V8 criterion	Local mathematical PASS; non-claim-bearing	independent replay pending
V9	localized centered-moment criterion for a positive L^2 eigenvalue	Proved	direct temporal operator-pencil argument
V9A	frozen `pulse_1` true profile satisfies the V9 criterion	Local mathematical PASS; non-claim-bearing	independent replay pending
V10	exact exclusion of classical stationary Turing onset and frozen-box separation	Proved	direct Fourier-symbol calculation
T2G	one global exact saddle chart and global winding alphabet	Proposed	not used
S1	temporal spectral or nonlinear stability	Deferred	separate problem
E1	calibrated experimental observation	Deferred	separate problem

5. Application proof locations

The reader-facing paper compresses the proof into four mathematical sections. The complete coordinate-level arguments remain in the versioned archive:

output	repository proof file
physical model, clocks, Hamiltonian	`van-der-pol/HAMILTONIAN_CHECK.md`; `van-der-pol/MODEL_AND_CENTRAL_CHART.md`
frozen core statement	`van-der-pol/CENTRAL_CORE_IMPORT.md`
central continuation	`van-der-pol/CENTRAL_CONTINUATION.md`
positive pole terminal	`van-der-pol/POSITIVE_POLE_FINITE_PART.md`
outer future-staying graph	`van-der-pol/OUTER_FUTURE_STAYING.md`
relative corner invariant manifold	`theory/RELATIVE_OVERFLOWING_NHIM.md`
central–outer matching	`van-der-pol/CENTRAL_OUTER_MATCHING.md`
algebraic finite parts	`van-der-pol/OUTER_ALGEBRAIC_FINITE_PART.md`
frozen return/coding statement	`van-der-pol/RETURN_EXIT_CODING_IMPORT.md`
conditional compact-family transfer	`theory/COMPACT_FAMILY_FIRST_HIT_THEOREM.md`
physical finite-atlas descent	`theory/FINITE_MARKED_ATLAS_DESCENT.md`
first events, actions, and PDE coding	`van-der-pol/TWO_END_RETURN_EXIT_AND_PDE.md`
arrow-by-arrow proof audit	`proof-audit/VDP_PUBLICATION_PROOF_AUDIT_2026-08-28.md`

6. Replay instructions and evidence level

Two replay statements must be kept distinct. The Zenodo archive includes a public replay summary with status `PASS` for the fixed-system paper's M1–M4 `main-theorem` profile. That replay concerns the four computer-assisted RFSN-II inputs only. It neither validates the companion's positive annulus nor constitutes an independent review of the proofs.

In a full Git clone of the application repository checked out at the release tag, change to the frozen `source/` directory and run

```
python3 validation/replay_all.py \
  --profile main-theorem \
  --dry-run \
  --report /tmp/rfsn-main-theorem-dry-run.json
```

checks the manifest and frozen input hashes. Its successful status is

`PASS-DRY-RUN-NO-CERTIFICATES-EXECUTED`.

By design, this is not a mathematical replay. The frozen script is fail-closed and invokes Git, so this command is not advertised for a GitHub source ZIP or for a standalone copy without `.git`. Its report field `git_commit` identifies the enclosing application checkout; the upstream frozen identity is the `upstream_commit` in `RELEASE_MANIFEST.json`.

A full replay uses the same command without `-dry-run`, supplies the CAPD source and `capd-config` paths, and follows the compiler and FILIB constraints in the frozen environment lock. The interval computations use outward-rounded CAPD/FILIB enclosures; see the frozen validation README and the software records in the source package. The package is included so that the source claim can be inspected and replayed. This release does not claim that a second machine replay was performed by the application project.

The application repository also contains an unfinished explicit positive-box validation programme under `validation/rigorous/`. Its aggregate status remains `INCONCLUSIVE`, `claim_bearing=false`. Its development environment lock is distinct from the frozen core environment above and is not evidence for the existential annulus in the paper.

7. Review and licensing status

The companion was subjected to mathematical seam audits, rendered-PDF checks, and a separate AI cold-reading task. These checks are not a substitute for an independent human expert review. A neutral expert packet is included beside the manuscript; no human reading is claimed until a named reader has actually returned a report.

The Zenodo fixed-system deposit contains its own paper and code license files. Those licenses do not retroactively license the distinct application repository or its frozen long snapshot. Neither of the latter contained a license file at the time of the application release; that snapshot therefore preserves inspection and citation access without inventing reuse rights. A future license decision for the application materials belongs to the repository owner and is not a mathematical claim of this supplement.

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